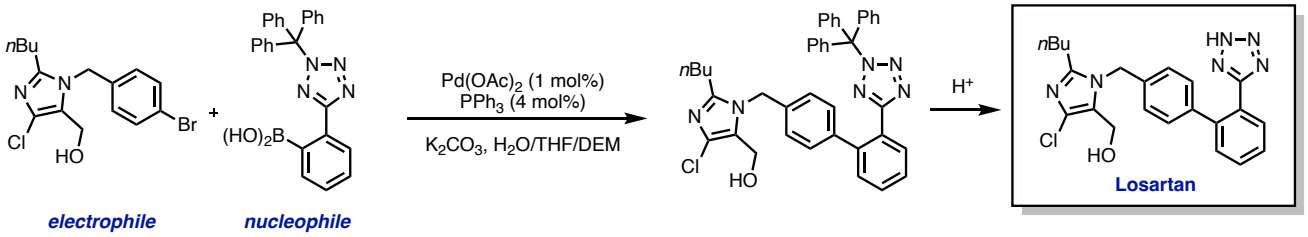
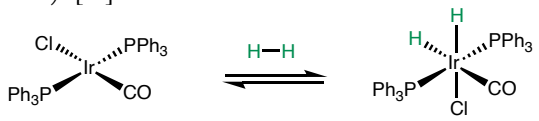
1. Indicate which coupling partner reacts as nucleophile and which one as electrophile in the following Suzuki coupling (*J. Org. Chem.* **1994**, *59*, 6391−6394) [★] [DEM, diethoxymethane]



The aryl bromide species undergoes oxidative addition onto low-valent Pd(0), thus acting as formal electrophile. The tetrazole-based boronic acid undergoes transmetalation with Pd(II) species, thus acting as formal nucleophile.

2. Suggest a possible spectroscopic technique to follow the oxidative addition of dihydrogen on Vaska's complex (*J. Am. Chem. Soc.* **1962**, *84*, 679–680). Explain why the observed kinetic isotope effect when D_2 is used is only $\frac{k_H}{k_D} = 1.22$ (*J. Am. Chem. Soc.* **1966**, *88*, 3511–3514). [★]

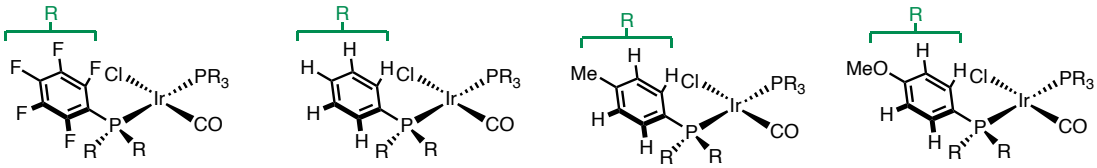


Both Ir−H and and C=O infrared (IR) resonances are very distinctive and can therefore be used to analyze the structural and electronic changes of Vaska's complex. The first stretching band appears at approx. 2100 cm^{-1}, while the carbonyl stretching at approx. 1970 cm^{-1}. The first resonance allows to determine whether H_2 undergoes oxidative addition (Ir^{I} to Ir^{III}) or there is simple side on coordination (donation of $\sigma_{H\text{-}H}$ to an empty *d* orbital and backdonation to $\sigma^*_{H\text{-}H}$) of H_2. The second resonance monitors the metal oxidation state and its electron-richness. This is because carbonyl ligands possess strong π-backbonding interaction with the metal, which is proportional to the electron richness of the former. Increased π-backbonding determines the weakening of the C=O bond, which results in lower stretching frequencies (*cf.* Tolman electronic parameter, see *J. Am. Chem. Soc.* **1970**, *92*, 2953–2956; *Chem. Soc. Rev.* **1972**, *1*, 337–353; *Chem. Rev.* **1977**, *77*, 313–348). For instance, *trans*-$[Ir^{I}(PPh_3)_2(CO)Cl]$ possesses $\upsilon_{CO} = 1967\ cm^{-1}$, while its oxidative addition complex $[Ir^{III}(PPh_3)_2(D_2)(CO)Cl]$ possesses $\upsilon_{CO} = 2034\ cm^{-1}$. This is consistent with a less electron-rich metal center, which reduced π-backbonding at the carbonyl ligand.
In order to account for the relatively small primary kinetic isotope effect, an early transition state (which resembles the reactants) was invoked. Computational studies by Goldman, Krogh-Jaspersen and co-workers (*J. Phys. Chem.* **1993**, *97*, 5890–5896) have proposed a more refined alternative theory. The kinetic isotope effect can be considered the contribution of three terms, namely mass-moment of inertia (MMI), EXC (vibrational excitation component) and differential zero-point energy component (EXP(ZPE)):

$$KIE = MMI \cdot EXC \cdot EXP(ZPE)$$

Since replacing hydrogen for deuterium has little effect on the mass of complexes, the MMI component is big and larger than one (5.56). The other two terms are inverse (<1) and therefore contribute to the overall small kinetic isotope effect. The strongly inverse EXC component likely arises from the number of low-frequency vibration in the oxidative addition complex, which are more populated in the case of D_2 *vs.* H_2. The inverse EXP(ZPE) is caused by the increased number of stretching and bending component in the product, whose frequency diminishes upon deuteration.

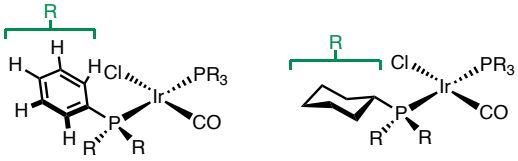
3. Rationalize the following bimolecular kinetic constants (k_2) for the for the oxidative addition of H_2 with the following Vaska's-type complexes (*Ann. N. Y. Acad. Sci.* **1971**, *172*, 546-562). [★]



k_2 = n.d. k_2 = 1.2 s^{-1} M^{-1} k_2 = 1.7 s^{-1} M^{-1} k_2 = 2.1 s^{-1} M^{-1}

When considering oxidative addition, since the metal oxidation state formally increases by 2 (Ir^{I} to Ir^{III} in this case) and two electrons are donated to the substrate, more electron-rich metals experience lower reaction barriers and faster kinetics. Phosphines are strongly donating ligands, and substitution with electron-donating groups causes increased donation to the metal center (*p*-Me and *p*-MeO). Pentafluorophenyl substituents reduce the electron-density at the metal, thus suppressing the oxidative addition process.

4. Explain the following seemingly contradictory data trends, where k_2 is the bimolecular kinetic constant for the for the oxidative addition of H_2 and υ_{CO} is the stretching frequency of the carbon monoxide ligand (*Ann. N. Y. Acad. Sci.* **1971**, *172*, 546-562). [★★]

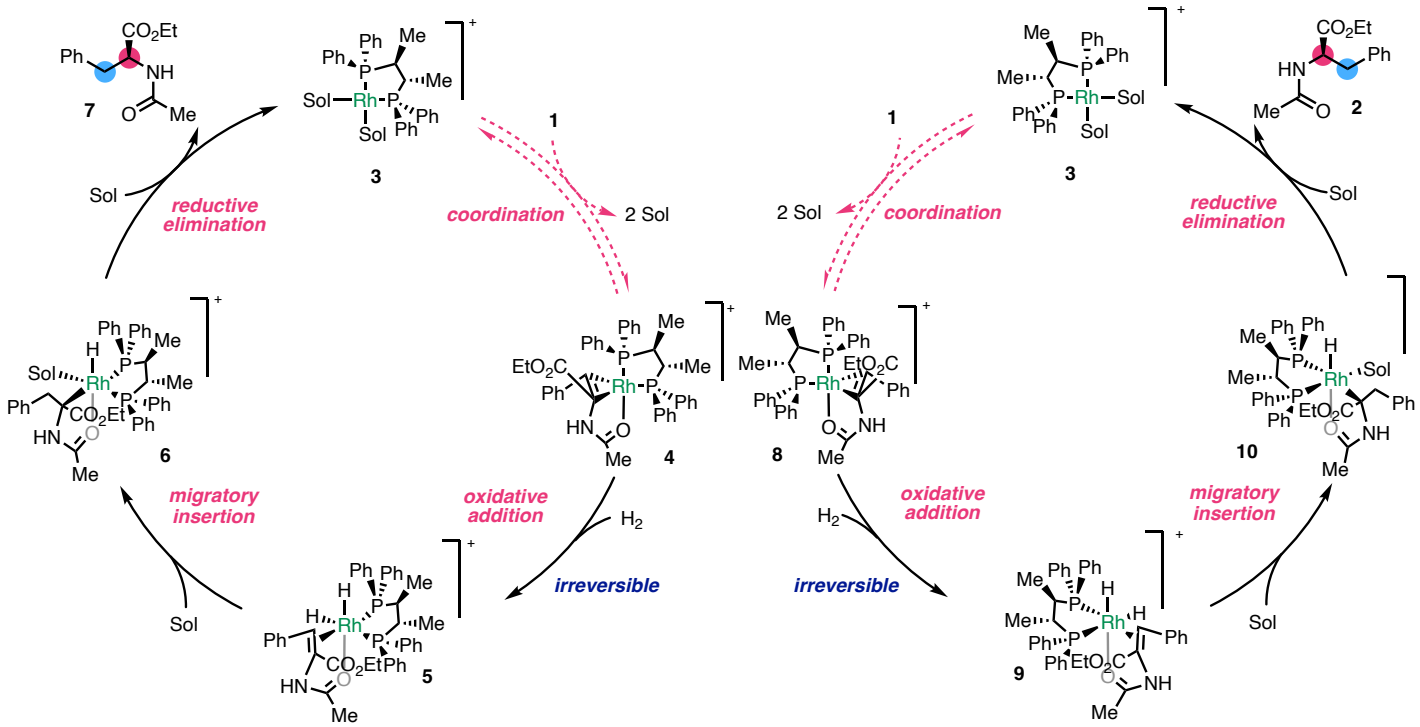


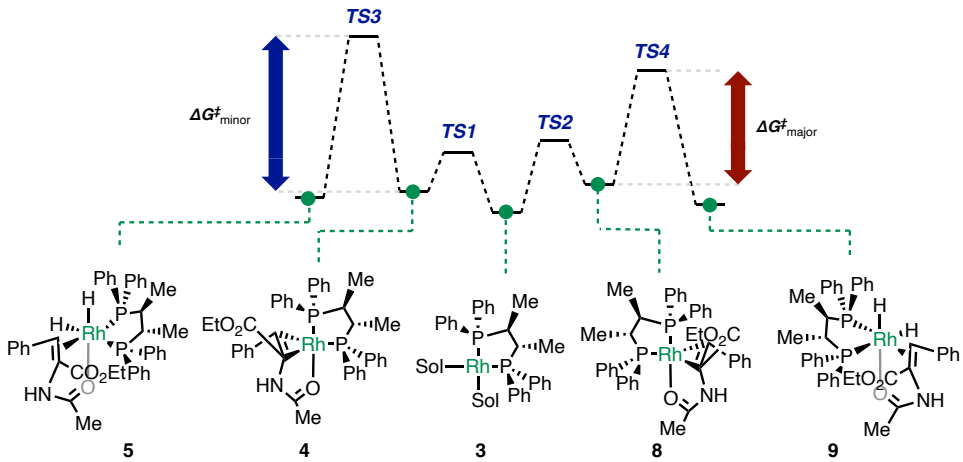
k_2 = 1.2 s^{-1} M^{-1} k_2 = 6.6 10^{-3} s^{-1} M^{-1}

υ_{CO} = 1954 cm^{-1} υ_{CO} = 1934 cm^{-1}

As previously mentioned, we can use carbonyl stretching frequencies to proxy the electron-richness of the metal center. Alkyl phosphines are more electron-donating than aryl phosphines, and coherently cause lower υ_{CO}. Nevertheless, the rates of oxidative addition are not solely due to the electronic features at the metal, but steric bulk plays a fundamental role. Tricyclohexyl phosphine is very sterically demanding (cone angle = 182°) compared to triphenylphosphine (cone angle = 145°) and cannot adopt a flat conformation to minimize the steric clash with the incoming species. In fact, Vaska himself realized that [Ir^{I}(*o*-tolyl-phosphine)$_2$(CO)Cl] does not undergo oxidative addition due to steric shielding of the metal center (*Inorg. Chem.* **1975**, *14*, 2669-2675).

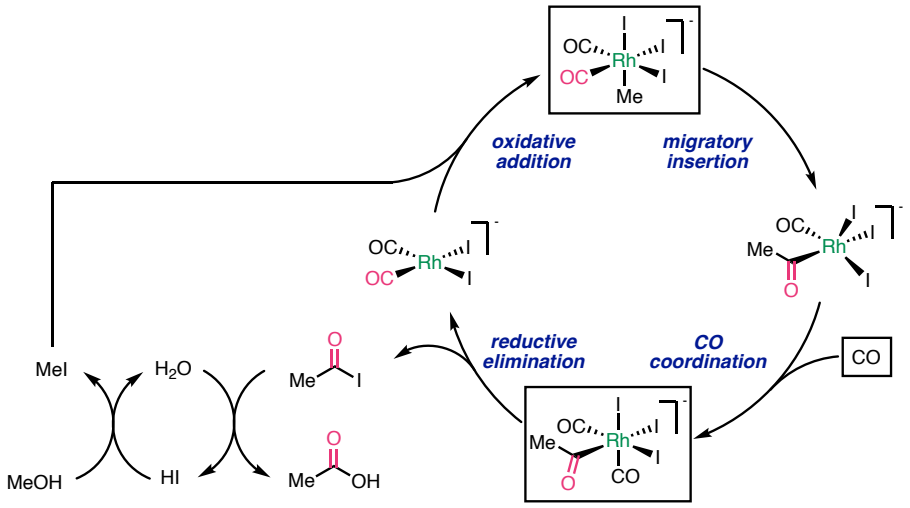
5. Halpern and co-workers studied the Rh-catalyzed asymmetric hydrogenation of prochiral acetoamidocynnamates (*J. Am. Chem. Soc.* **1980**, *102*, 5952-5954) and found the following: (a) The authors observed in solution and isolated a single diastereoisomer (**4**) from alkene coordination, but if the commonly accepted mechanism is followed (migratory insertion, *syn*-reductive elimination) the opposite enantiomer (**7**) of the observed product (**2**) would be generated; (b) The rate law reaction is $-\frac{d[\mathrm{H_2}]}{dt} = k_{ox.add.}[\mathbf{4}][\mathrm{H_2}]$, which tells that the H_2 oxidative addition is the rate determining step. Rationalize the stereochemical outcome of the reaction. (This system operates analogously to the DIPAMP-based Knowles hydrogenation, see *Angew. Chem. Int. Ed.* **2002**, *41*, 1998-2007) [★★]





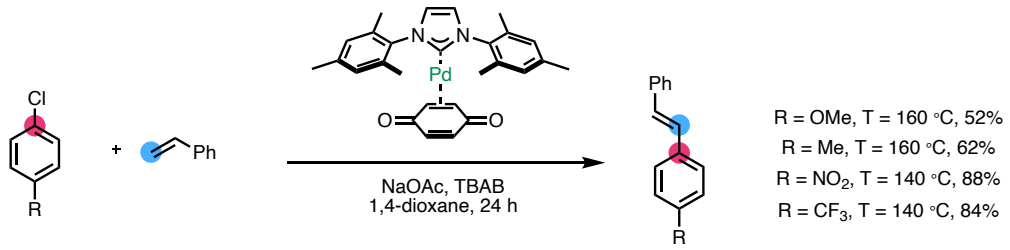
The most likely scenario, corroborated by in-depth mechanistic experiments, is that complex **4** is preferentially formed because of its higher stability compared to **8**, but it reacts slower in the following steps. More specifically, the reaction operates under Curtin-Hammett regime, whereby initial alkene coordination is fast and reversible, and favors the more stable isomer. The reaction selectivity is decided by the relative rate between competitive oxidative addition reactions (first irreversible reaction), which is proportional to $\Delta\Delta G^{\ddagger}$. In this case, oxidative addition of H_2 via TS4 is more favorable than through diastereomeric TS3, and this control the overall enantioselectivity. Unsurprisingly, the kinetic equation for the whole process is the bimolecular rate law of the oxidative addition step ($-\frac{d[H_2]}{dt} = k_{ox.add.}[\mathbf{4}][H_2]$)

6. Complete the proposed catalytic cycle of the Monsanto process (acetic acid synthesis, for an overview: Hartwig, J. F. Organotransition Metal Chemistry, from Bonding to Catalysis; University Science Books: New York, 2010). Explain why for rhodium (Monsanto process) the oxidative addition is rate-determining, while for the BP-Cativa process (based on iridium) is the migratory insertion step. [★★]



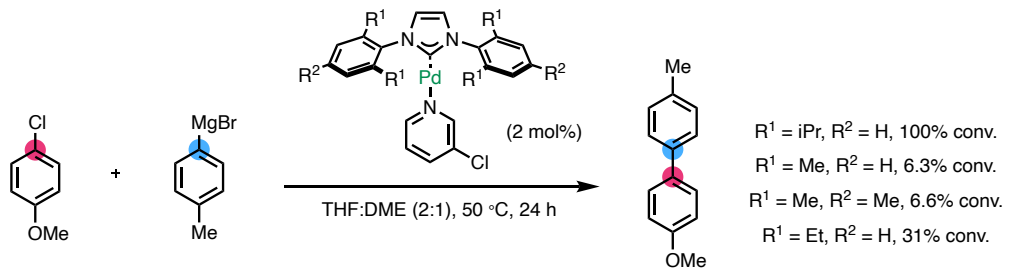
The Monsanto process involves initial oxidative addition of methyl iodide (generated *in situ* from methanol) with anionic $[Rh(I)_2(CO)_2]^-$ in S_N2 fashion to give octahedral complex $[Rh(Me)(I)_3(CO)_2]^-$. Migratory insertion is followed by carbon monoxide coordination to $[Rh(MeCO)(I)_3(CO)_2]^-$, which undergoes reductive elimination to give acetyl iodide.
Since oxidative addition increases the ligand count by two, second row transition metals such as rhodium (which have smaller radii) suffer from increased steric crowding, making the process more energetically demanding. Conversely, third row transition metal have larger coordination space to accommodate additional ligands. Furthermore, the increasingly soft character of the metal, due to more diffused orbitals, renders the center more nucleophilic. In the case of the migratory insertion step, metal-alkyl bonds are stronger for third row transition metals than second row ones. This results in an increased energy barrier for the bond-breaking migratory insertion for the BP-Cativa process.

7. In 2002, Umicore patented (World Patent 2004, 014, 550, 2003) the following Pd-carbene pre-catalyst for the Mizokori-Heck coupling of aryl chlorides. Explain the ligand role and hypothesize the reason behind the yields trend (higher for aryl chloride with electron-withdrawing substituents and required temperature). [TBAB, tetrabutylammonium bromide] [★]



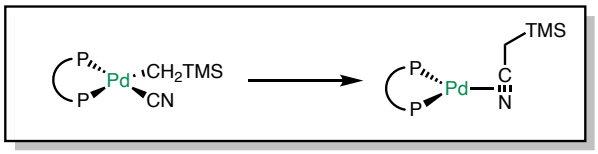
Aryl chlorides are significantly less prone to undergo oxidative addition, compared to iodides and bromides. Special ligand systems are strong electron donors (*e.g.* N-heterocyclic carbenes) enhance the rate of oxidative addition, since electron-rich and low oxidation state metals undergo more facile oxidative addition. Controlled steric bulk around the metal (in this case through the two flanking mesityl groups) can liberate required coordination space for the oxidative addition to happen. Despite being somewhat counterintuitive (reductive elimination, the opposite process, is facilitated by increased steric bulk around the metal), the coordination space liberation effect can override the increased steric bulk if particularly strongly binding ancillary ligands are present.
The yield and temperature trends can be rationalized considering that electron-poor arenes undergo oxidative addition more easily than electron-rich ones. This is due to the increased tendency to accept two electrons from the metal upon oxidative addition. An analogous point of view emphasized the more favorable formation of Pd(0)-arene pre-reactive complex, which renders the following oxidative addition more energetically feasible (early transition state, see: *Organometallics* **2007**, *26*, 550-553)

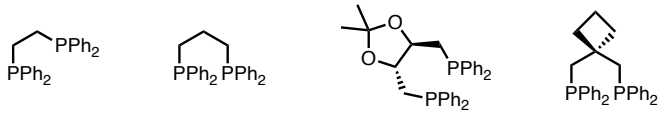
8. Explain the following trends in the Kumada coupling using PEPPSI-type ligands (*Chem. Eur. J.* **2010**, *16*, 10844-10853). [★]



As outlined in the previous answers, two effects can contribute to the more or less favorable oxidative addition: (a) electronic effects, with more electron rich metal centers being more favorable; (b) Steric bulk around the metal, which could either make the oxidative addition less facile or enable that *via* enhanced ligand dissociation. In the present report, the author realized that σ-donating ability of the ligands is comparable across the board. The main difference is the substantially different steric demand, with more sterically bulky ligands giving the best results. Since poorly donating 3-chloropyridine is used as easily on-off ancillary ligand, it is not believed that steric bulk plays a major role in liberating coordination space for the electrophile addition. The authors suggest that reductive elimination, which is facilitated by bulky ligands, is responsible for the observed trend. The ligand imposes steric pressure around the metal center, which brings the two aryl ligands together and leads to lower energy barriers (in addition to the strain release effect).

9. Marcone and Moloy (*J. Am. Chem. Soc.* **1998**, *120*, 8527-8528) studied the rate of reductive elimination from Pd(II) complexes, and observed the following rates. Hypothesize what causes the following rate trends and justify the difference between dppp and the corresponding cyclobutylated version. [★★]

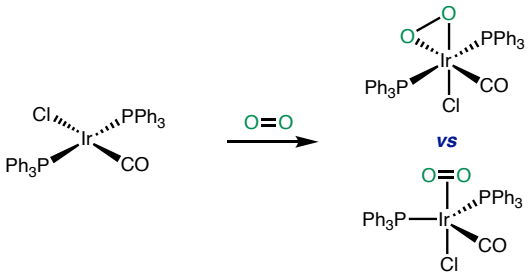




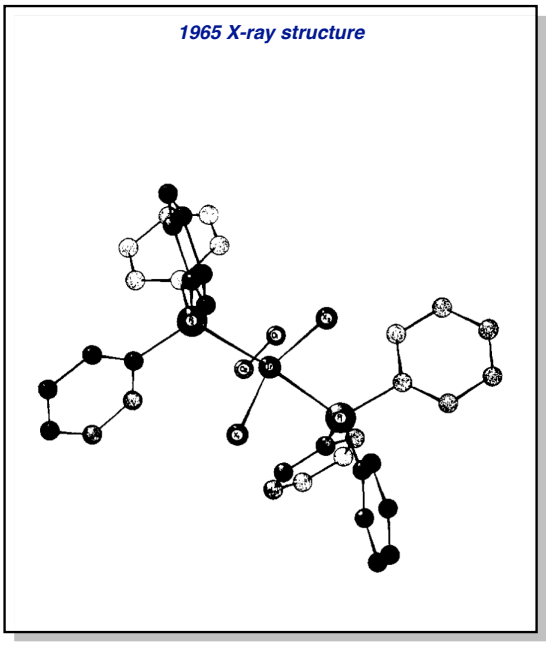
bite angle	86°	91°	102°	n.d.
k (s⁻¹)	$2.1\ 10^{-6}$	$5.0\ 10^{-5}$	$1.0\ 10^{-2}$	$7.4\ 10^{-5}$

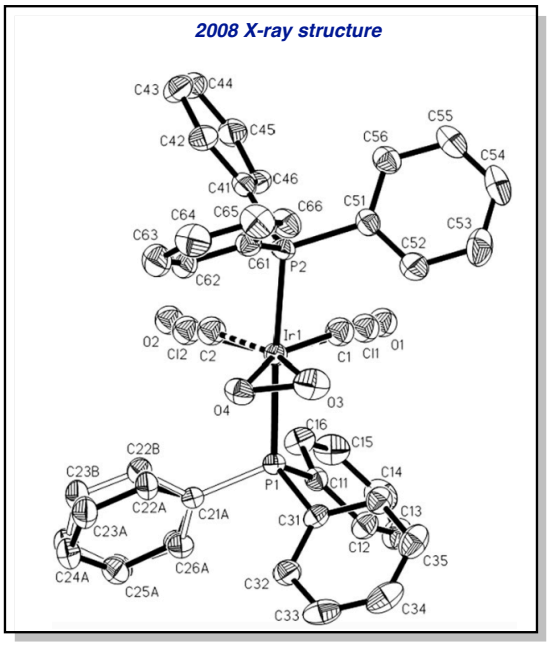
We observe a correlation between increasing bite angle (see: *Chem. Rev.* **2000**, *100*, 2741-2770) and reaction kinetic. Analogously to steric bulk, large bite angles force substituents to come in close proximity, thus rendering the transition state more energetically feasible. The rate difference between dppp and its cyclobutylated version likely arises from a Thorpe-Ingold-like effect. Cyclization creates a more geometrically constrained environment around the metal (despite the probably lower bite angle, *cf. Angew. Chem. Int. Ed.* **2018**, *57*, 9103-9107), which results in more favorable reductive elimination.

10. Suggest a method to distinguish the oxidative addition product of O_2 from the simple coordination (analogous to an olefin Dewar-Chatt-Duncanson model) (*Science* **1964**, *145*, 920-921 and *Organometallics* **2015**, *34*, 4784-4795). [★]



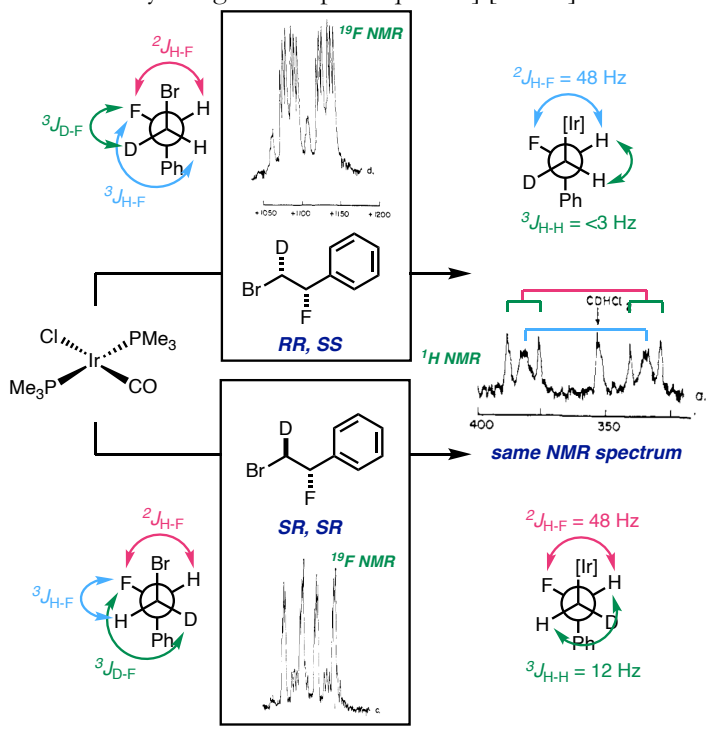
C=O infrared (IR) resonances are very distinctive and can therefore be used to analyze the structural and electronic changes of Vaska's complex. The carbonyl stretching for Ir^{I} species is located at approx. 1970 cm^{-1} and can be used to monitor the metal oxidation state. This is because carbonyl ligands possess strong π-backbonding interaction with the metal, which is proportional to the electron richness of the former. Increased π-backbonding determines the weakening of the C=O bond, which results in lower stretching frequencies (*cf.* Tolman electronic parameter, see *J. Am. Chem. Soc.* **1970**, *92*, 2953–2956; *Chem. Soc. Rev.* **1972**, *1*, 337–353; *Chem. Rev.* **1977**, *77*, 313–348). Two alternative views of the interaction between oxygen and Vaska's complex are either a π adduct of neutral dioxygen or a real oxidative addition η^2-peroxo complex. The $\upsilon_{CO} = 2015\ cm^{-1}$ reflect a significant withdrawal of electron density from the metal center but does not allow us to assess which one of the structure describes better the binding interaction. υ_{OO} could therefore be used: for $[Ir^{I}(PPh_3)_2(O_2)(CO)Cl]$ the value of $\upsilon_{OO} = 850\ cm^{-1}$, which is comparable to peroxide O_2^{2-} ($\upsilon_{OO} = 820\ cm^{-1}$). On the other hand, the values for superoxide O_2^{-} ($\upsilon_{OO} = 1145\ cm^{-1}$,) and O_2 ($\upsilon_{OO} = 1556\ cm^{-1}$,) are far away from the experimental value, thus corroborating the η^2-peroxo complex structure.
X-ray crystal structure can allow a direct insight into the bond lengths and molecular structure. La Placa and Ibers solved the crystal structure for the complex back in 1965 (*J. Am. Chem. Soc.* **1965**, *87*, 2581-2586), but the O−O bond distance was erroneously reported (1.30 Å, comparable to superoxide distance of 1.33 Å). In 2008, a new measurement was conducted and gave a value of 1.47 Å (comparable to peroxide, 1.45 Å).





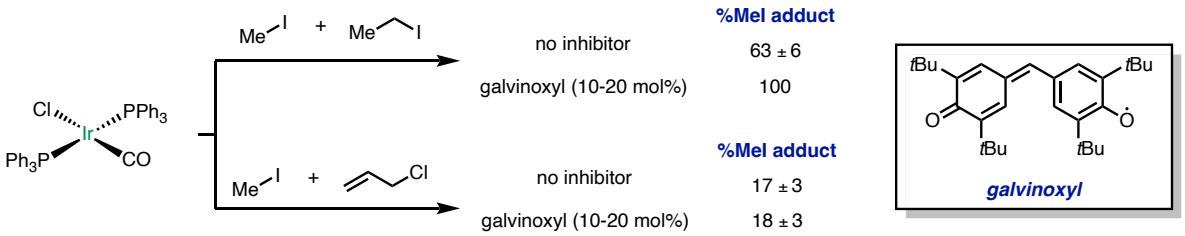
11. Osborn and co-workers (*J. Am. Chem. Soc.* **1972**, *94*, 4043-4044) studied the oxidative addition of alkyl bromide with Vaska's-type complex *trans*-$[Ir(PMe_3)_2(CO)Cl]$. The authors synthesized either the *RR,SS* or *RS,SR* dyad of a fluoro-deutero-bromide, which could allow facile NMR study of the oxidative addition complex. Independent from the starting dyad, the same NMR spectrum containing resonance for both diastereoisomers was observed. Explain which oxidative

addition mechanism is operative. [The different coupling constants, which allow unequivocal identification of the different species, can be estimated by using the Karplus equation] [★★★]



Three main oxidative addition mechanism, which lead to different stereochemical considerations, can be conceived: (1) Concerted *syn*-oxidative addition (*e.g.* aryl halides), which causes retention of configuration; (2) S_N2-type oxidative addition (*e.g.* Rh-catalyzed Monsanto process), which causes inversion of configuration; (3) Single-electron transfer leading to radical formation, which recombines to create Ir(III)-alkyl species, will cause epimerization (radicals are generally not conformationally stable). Assuming that the produced Ir(III)-alkyl complexes are conformationally stable (*i.e.* no dissociation to generate configurationally unstable radical species occurs), we observe the third scenario. Particularly, the NMR spectrum shows an approx. 1:1 mixture of *cis* and *trans* isomers. The first is characterized by large geminal $^2J_{H-F}$ coupling and negligible $^3J_{H-H}$ (the Karplus equation predicts at approx. 85° dihedral angle), while the second shows large geminal $^2J_{H-F}$ coupling and substantial $^3J_{H-H}$ (the Karplus equation predicts at approx. 180° dihedral angle).

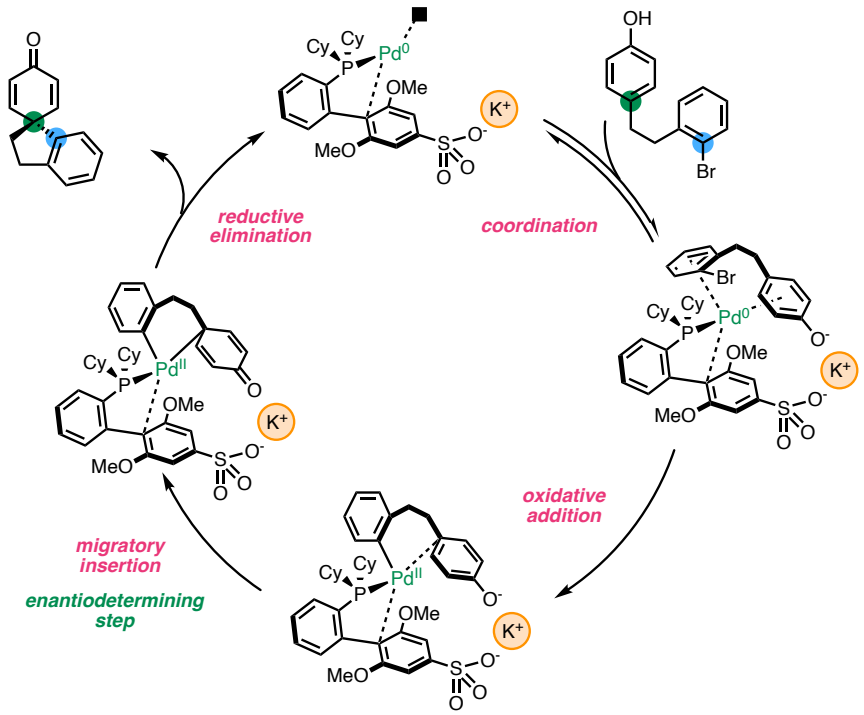
12. Labinger, Osborn and co-workers (*Inorg. Chem.* **1980**, *19*, 3236-3243) evaluated the impact of a radical inhibitor (galvinoxyl) on Vaska's-type complex *trans*-[Ir(PMe$_3$)$_2$(CO)Cl] on the competitive oxidative addition between methyl iodide and different electrophiles. Explain what the following data suggest about the mechanism of oxidative addition for ethyl iodide and allyl chloride and suggest an additional experiment to determine the exact oxidative addition mechanism for allyl chloride. [★★]



The underlying hypothesis is that radical inhibitors such as galvinoxyl can only influence a radical pathway by scavenging radical species. When comparing methyl and ethyl iodide, the predominant but not exclusive adduct is the methyl derivative. Upon

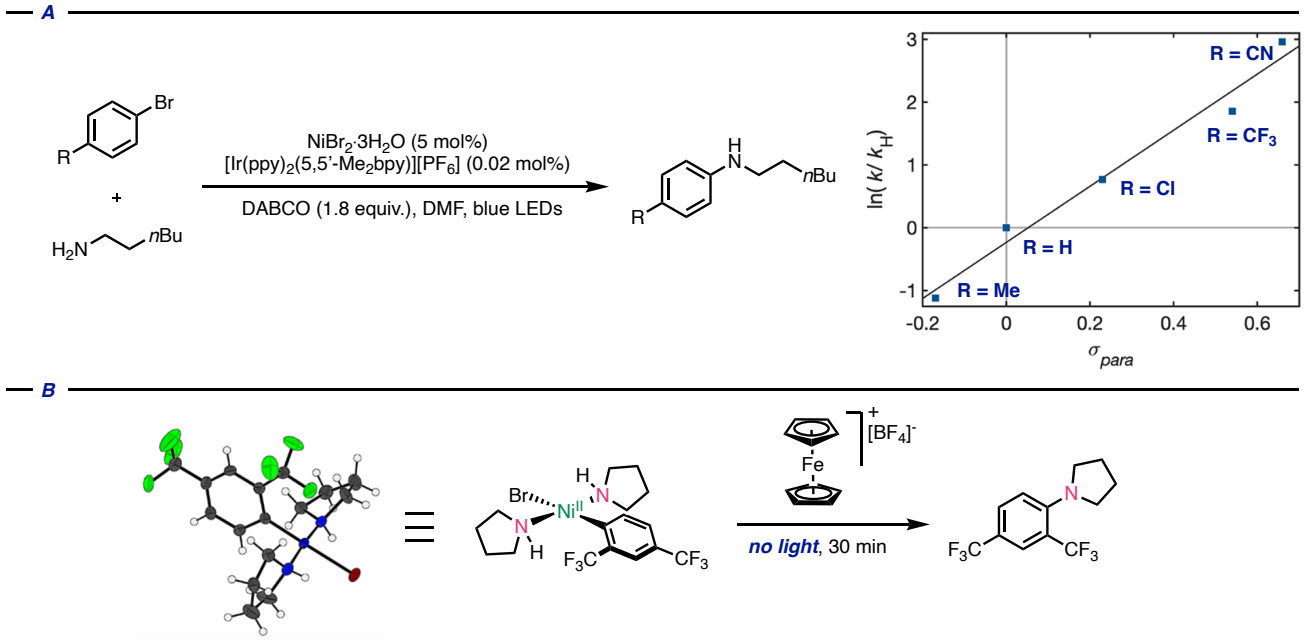
addition of galvinoxyl, the formation of the ethyl adduct is completely suppressing, thus suggesting that ethyl iodide undergoes oxidative addition *via* radical mechanism. When comparing methyl iodide and allyl bromide, no difference between the reaction conducted in absence or presence or galvinoxyl can be detected. This suggests that the neither the oxidative addition of methyl iodide, nor of allyl bromide, occur via radical formation. Considering the high activation of allyl halide substrates towards S_N2-type reactions, we can speculate that this mechanism could be operative, but additional evidence is required to rule out a concerted *syn*-pathway. A possible strategy could be judging whether enantiopure substituted allyl halides undergo inversion (S_N2-type) or retention (concerted *syn*-pathway) of configuration. In case of S_N2'-type reactions, removal of the stereocenter render the approach unfeasible.

13. Phipps and co-workers (*J. Am. Chem. Soc.* **2023**, *145,* 25553-25558) reported the following enantioselective dearomative cyclization of phenols. Hypothesize the role of the ligand and draw a possible catalytic cycle. [★★]



The ligands represent a modification of the widely-applied Buchwald ligands and it's decorated with a sulfonate moiety. We can hypothesize that the ligands operates *via* a three-fold role: (a) activate the system towards the cross-coupling reaction (electron-rich, hemilabile phosphine which can control the steric bulk around the metal); (b) activates the phenol moiety towards the formal dearomative insertion at the *para*-position; (c) its coordination renders the transition states more compact, allowing to enforce high levels of enantioselectivity. The mechanistic scenario is the same as originally outlined by Buchwald in 2011 (*J. Am. Chem. Soc.* **2011**, *133,* 9282-9285), with the phenolate palladation step being selectivity determining: initial coordination of the aryl bromide is followed by oxidative addition to give Pd^{II}-aryl complexes. At this stage, both the phenolate and the sulfonyl moiety are electrostatically interacting with the potassium cation, thus locking the conformation of the pendant substrate. This enforces enantioselectivity in the migratory insertion onto the Pd^{II}-aryl intermediate, and it's followed by *syn*-reductive elimination to generate the dearomatized spirocyclic product. The authors performed the experiment with TMS-protected phenol and observed analogous reaction outcome, thus disproving the involvement of hydrogen-bond interaction (no proton is present). The replacement of potassium cations for quaternary ammonium salts proved highly detrimental to both yield and enantioselectivity, thus suggesting that the size of cations is crucial for the desired aggregation. When an appropriately-sized crown-ether (15-crown-5 or 16-crown-6) was added, enantioselective erosion was observed, further hinting that the cation enhances complex organization.

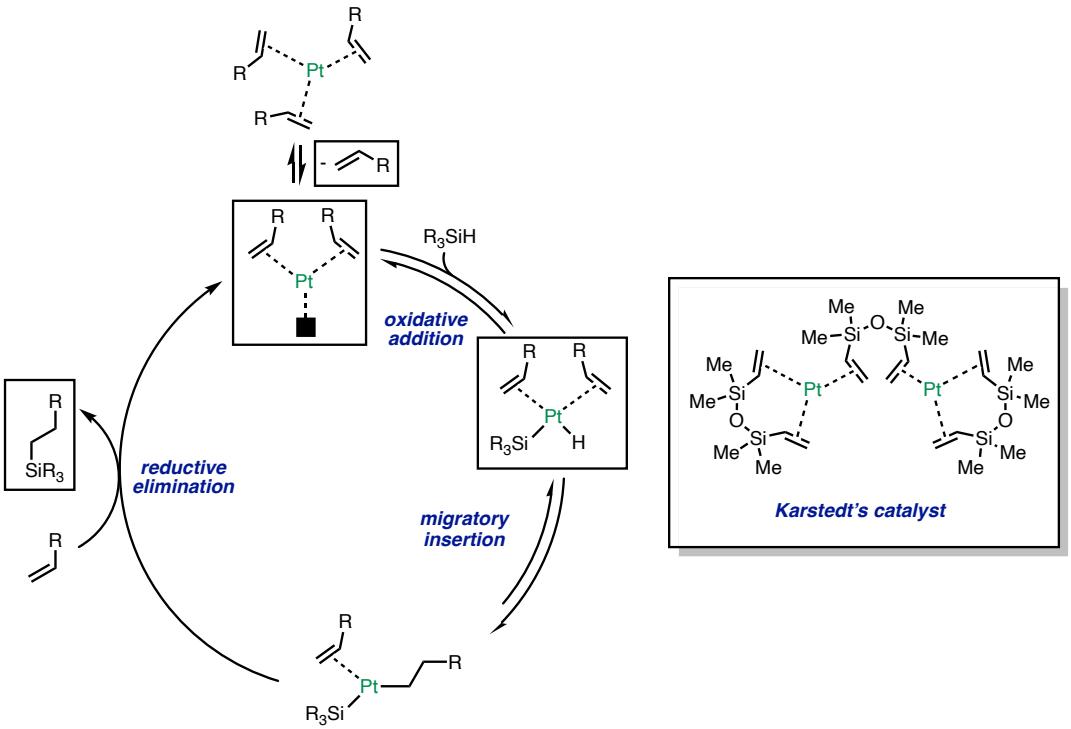
14. In 2020, MacMillan and co-workers studied a nickel metallaphotoredox catalyzed Buchwald-Hartwig-type coupling reaction (*J. Am. Chem. Soc.* **2020**, *142*, 15830-15841) and observed the following: (A) A positive *para*-Hammett correlation; (B) Upon single-electron oxidation of an intermediate Ni^{II}(aryl)(Br)(amine)$_2$ complex with ferrocenyl tetrafluoroborate (a single-electron oxidizing agent), quick product formation could be observed. Explain what these two observations suggest about oxidative addition and reductive elimination. [★★]



The reaction rate increases with electron-withdrawing groups, which is in accordance to the notion that more electron-poor aryl halides have lower energy barriers for oxidative addition. The strong positive Hammett correlation (ρ= +4.5) indicates that strong negative charge buildup occurs at the oxidative addition transition state, and EWG stabilize this charge buildup by inductive and mesomeric effect (*Chem. Rev.* **1991**, *91*, 165-195).

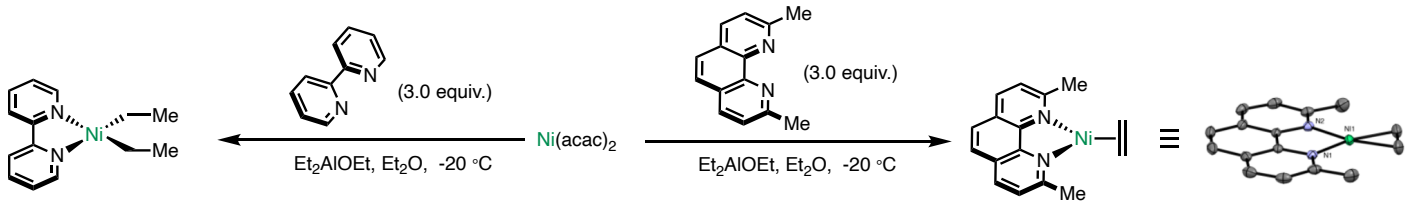
The isolation of Ni(II)-aryl-bisamine complexes shows that reductive elimination from Ni(II) complexes in generally unfavorable. Upon single-electron oxidation of Ni(II) to Ni(III) using $[Fc]^+[BF_4]^-$, the complex undergoes fast reductive elimination, due to the electron-poor character of the metal center. This suggests that Ni(III) species are competent in the reductive elimination step to generate the aminated product. The authors submitted a similar Ni(II)-aryl-trisamine complex to irradiation in presence of the photocatalyst and observed reductive elimination, thus hinting that the photocatalyst can induce single-electron oxidation and reductive elimination.

15. Olefin hydrosilylation using platinium complexes (*e.g.* Kartedt's catalyst represents an industrially important reaction to generate silicon-contaning species. Complete the mechanism below. [★] (For an example of mechanistic investigation, see *J. Am. Chem. Soc.* **1998**, *121*, 3693-3703).



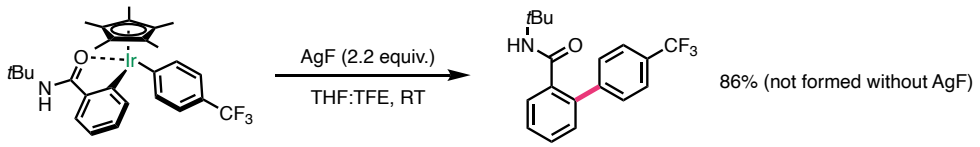
Pre-catalytic species usually involved olefin three-coordinated platinum(0) species, that must undergo ligand dissociation to generate a vacancy for oxidative addition to happen. Oxidative addition is usually believed to be reversible, and generates Pt(II)-hydride complexes. Hydrogen migratory insertion (Chalk-Harrod mechanism) occurs, but in specific situation silyl-migratory insertion (modified Chalk-Harrod mechanism) was observed (see for instance: *Organometallics* **1998**, *17*, 2510-2523). Irreversible reductive elimination generates the C−Si bond to deliver the product.

16. Martin and co-workers (*Organometallics* **2022**, *41*, 2662-2667), in the attempt to synthesize Ni^{II}-dialkyl complexes, observed the following behavior. Rationalize the effect of the two different ligands. [★★]



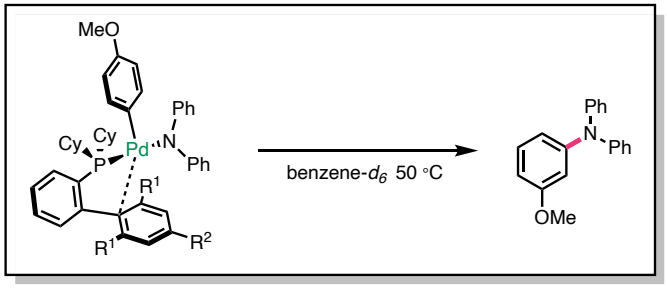
As shown by the stability of $[Ni(bpy)(Et)_2]$, Ni(II) complexes are less susceptible to *β*-hydride elimination when compared to the corresponding Pd(II) complexes. This is due to the metal small size, which imposes a prohibitive torsional barrier to overcome to achieve a reactive conformation ($\sigma_{C\text{-}H}$ and an empty nickel orbital should be aligned to achieve overlap). In the case of dimethylphenanthroline, *β*-hydride elimination (followed by reductive elimination) spontaneously occurs to generate the corresponding Ni(0) complex. This is likely due to the increased steric bulk around the metal, which promotes the formation of the less sterically demanding tricoordinate complex. Indeed, 2,9-disubstituted bispyridines are widely used in nickel-catalyzed "chain-walking processes" (see ref. 27 in paper), where *β*-hydride elimination/reinsertion steps are required to ensure migration along the chain.

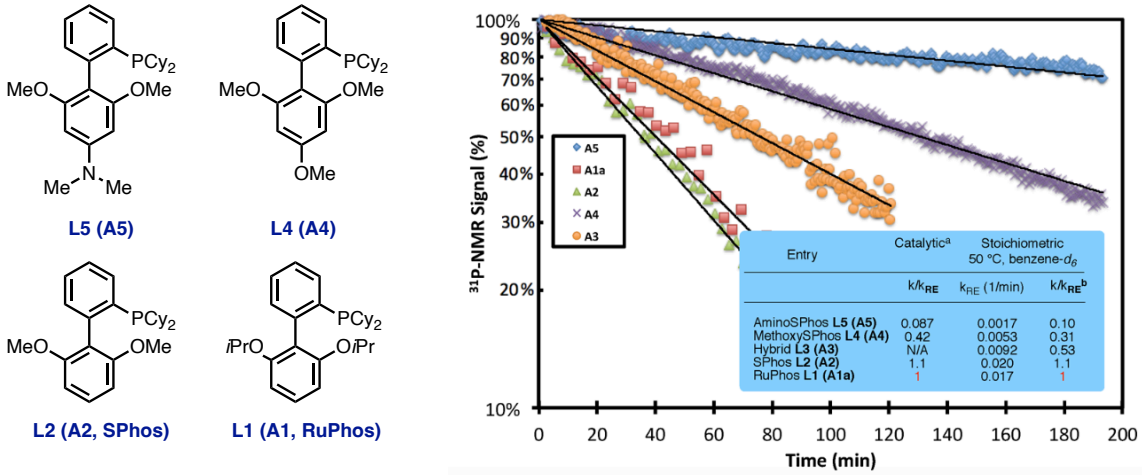
17. Chang and co-workers (*Nat. Chem.* **2018**, *10*, 218-224) reported that silver(I) oxidizing agents (*e.g.* AgF) are required to trigger reductive elimination in cyclometalated Ir(III) complexes. Rationalize the mechanism and provide an explanation to the observation. [★★]



This reaction shares similarities with the case of MacMillan and co-workers (*J. Am. Chem. Soc.* **2020**, *142*, 15830-15841). The intermediate Ir(III) intermediate does not spontaneously undergo reductive elimination. Oxidation of the metal center generates Ir(IV) or Ir(V) species, that have increased driving force for reductive elimination. The authors calculated that reductive elimination from Ir(III) possesses an energy barrier of $\Delta G^{\ddagger}$ = 34.4 kcal mol^{-1}, while for Ir(IV) is $\Delta G^{\ddagger}$ = 15.7 kcal mol^{-1} and Ir(V) $\Delta G^{\ddagger}$ = 4.5 kcal mol^{-1}. Interestingly, the Ir(III)→Ir(I) reductive elimination is endoergonic, and according to the Hammond postulate it is characterized by a late TS. On the other hand, , the Ir(IV)→Ir(II) and Ir(V)→Ir(III) reductive eliminations are exoergonic and characterized by an early TS.

18. Buchwald and co-workers (*J. Am. Chem. Soc.* **2016**, *138*, 12486-12493) studied the rate of reductive elimination in the eponymous coupling reaction and observed that more electron-rich ligands such as **L4-L5** caused slower rates than **L1-L2**. Provide an explanation for the observation. [★★]

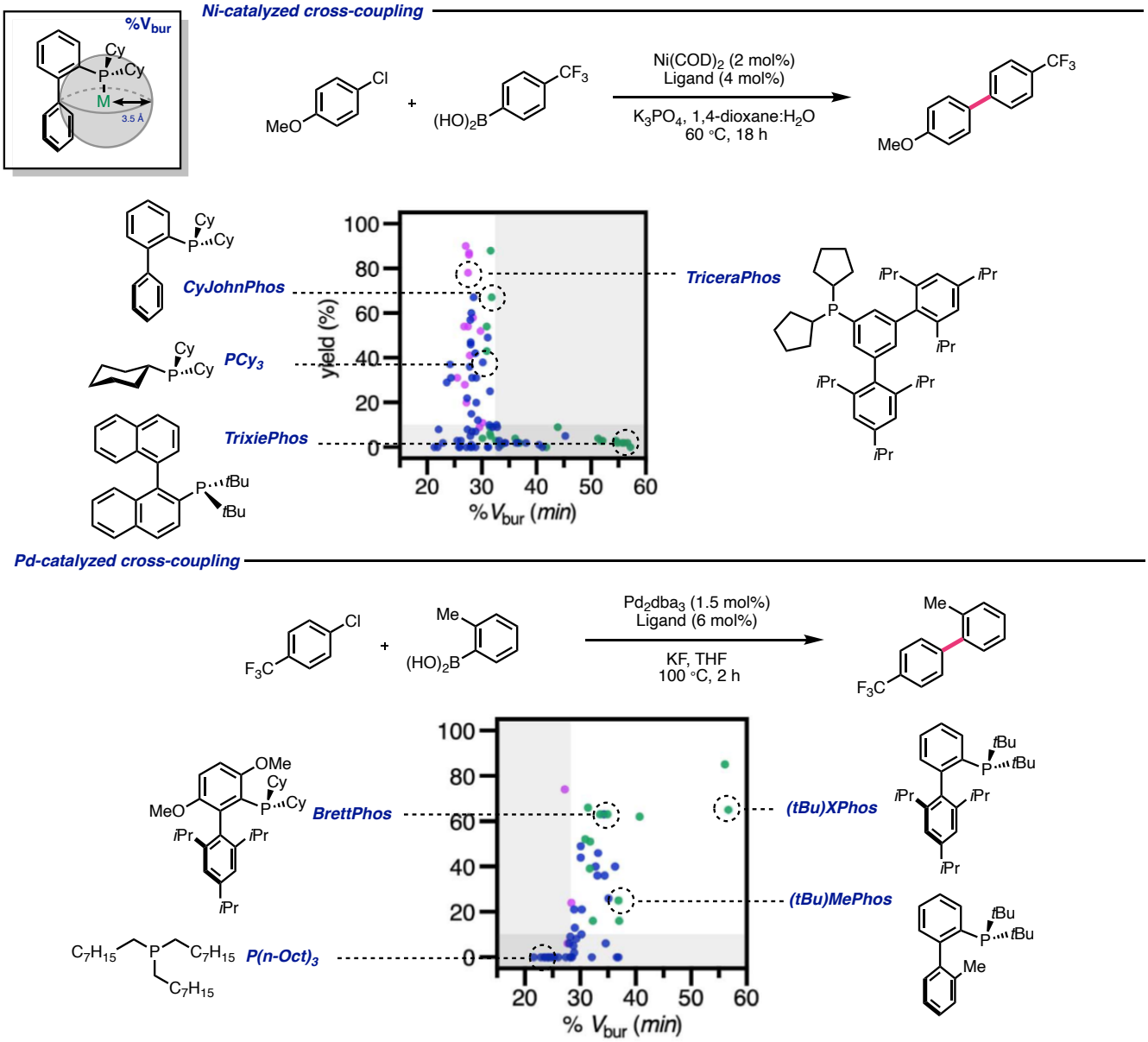




Entry	Catalytic[a] k/k_{RE}	Stoichiometric 50 °C, benzene-d_6 k_{RE} (1/min)	Stoichiometric k/k_{RE}[b]
AminoSPhos **L5 (A5)**	0.087	0.0017	0.10
MethoxySPhos **L4 (A4)**	0.42	0.0053	0.31
Hybrid **L3 (A3)**	N/A	0.0092	0.53
SPhos **L2 (A2)**	1.1	0.020	1.1
RuPhos **L1 (A1a)**	1	0.017	1

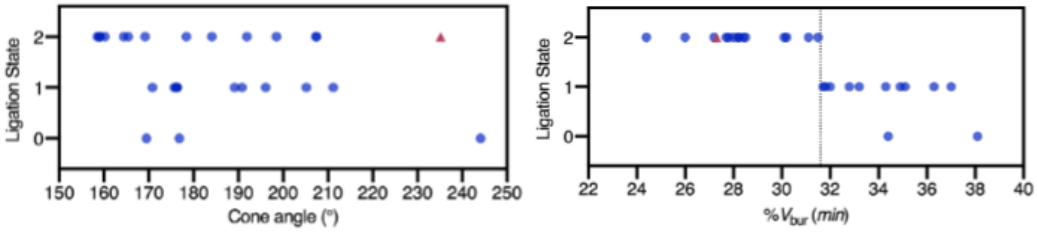
The rate of reductive elimination between SPhos and Ruphos is very similar, which likely indicates that the increased steric demand of RuPhos does not significantly impact the rate of the process. When electron-donating substituents are added (**L4-L5**), the reaction rate markedly decreases. Electro-rich ligands are believed to coordinate more strongly in the amido complex, reducing the rate of reductive elimination. More electron-rich metals show lower elimination rate, since their tendency to undergo formal two-electron reduction is lower.

19. Sigman, Doyle and co-workers (*Science* **2021**, *374*, 301-308) observed that nickel-catalyzed Suzuki coupling reactions with different ligands proceed when $\%V_{bur}$ (min) (the minimal energetically accessible $\%V_{bur}$, obtained from the data driven library *kraken*: *J. Am. Chem. Soc.* **2022**, *144*, 1205-1217) and the $\%V_{bur}$ (Boltz) (the average $\%V_{bur}$ obtained from the Boltzmann distribution), rather than the ligand cone angle. On the other hand, the authors observed that palladium-catalyzed Suzuki coupling reactions with different ligands proceed when $\%V_{bur}$ (min) is above a certain threshold Rationalize the experimental observations and compare the two catalytic systems. [$\%V_{bur}$ is the volume percentage occupied by the ligand within a sphere of 3.5 Å around the metal, see: *Chem. Commun.* **2010**, *46*, 841-846] [★★★]



The sharp cliff between successful or unsuccessful reaction is suggestive of a significant binary modification in the mechanism, that can render the catalytic cycle competent or generate an inactive species. Since monoligated species (L_1M) are known to be crucial for palladium-catalyzed cross-coupling reaction to proceed (see, for instance: *Acc. Chem. Res.* **2008**, *41*, 1461-1473), the authors believe that $\%V_{bur}$ (min) represents a parameter that acts as proxy of the steric environment around the metal, which ultimately influences the ligation at the metal. Indeed, NMR analysis revealed that $\%V_{bur}$ (min) correlated well with the tendency to form L_1Ni or L_2Ni species, while the cone angle had no significant influence. In the case of nickel, the cutoff at approximately

32% %V_{bur} (min) indicates that below this value L_2Ni species, which are reactive, are formed. Above this limit, unreactive monoligated species are preferentially formed. Again, no correlation with the cone angle (see image) was observed, while very good agreement with the hypothesis was observed for %V_{bur} (min).



Very large phosphine like Triceraphos, despite extreme cone angles (198.5°, Boltzmann), impart good reactivity, due to the limited %V_{bur} (min) (27.8%) which suggests the presence of active bis-ligated species. Differently, TrixiePhos possesses an extreme cone angle (213.3°, Boltzmann), but also elevated %V_{bur} (min) (57.2%) and is therefore unreactive (L_1Ni).
The case of palladium is reversed, since L_1Pd species are catalytically active and L_2Pd are not (with exceptions, see for instance the selective oxidative addition of chlorides over triflates, *Nat. Chem.* **2016**, *8*, 610-617). Depending on the requirement of the reaction different thresholds were found, again dependent on the tendence to form monoligated or bis-ligated species. Interestingly, the authors revealed two literature datasets where steric bulk (%V_{bur} (min)) thresholds don't seem to exist (*Chem. Eur. J.* **2001**, *7*, 2908-2915; *J. Am. Chem. Soc.* **2012**, *134*, 13765-13772).
Taken together, these data reveal the strikingly different ligands requirements for nickel and palladium-catalyzed reactions and the ability to gain mechanistically relevant information by analyzing the reaction behavior upon changing the ligand system.